Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

Enjoyable Lite

a. Team name: **ables.com.my**
b. Team leader name: **Lee Wei Song**
c. Problem Statement: **Sustainability & Social Impact: accessibility for blind and low-vision viewers**

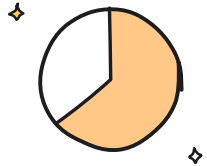


Brief about the idea



27 million

blind people live in Asia-Pacific, 6 in 10 worldwide



64%

of the world's vision loss is in Asia-Pacific



US$15–75

a minute for professional description, so almost no video has it

Enjoyable Lite

Paste **any YouTube link**. **Gemini** watches the video, finds the pauses in the dialogue and writes a short description for each one.

A natural voice speaks each line inside its pause while the video plays, in 33 languages. Speech in another language can be **dubbed** into yours.

No login. Keyboard and screen-reader first. Every description is saved, so the next viewer hears it instantly.

Paste a link › **Gemini writes** › **You hear it**

Sources: IAPB Vision Atlas (GBD 2020); 3Play Media

Opportunities

How it differs from what exists, and what only it does



	What it covers	In the pauses	Languages	Cost per minute
Studio description	Selected films and shows	✓	A few	US$15–75
Creator-made description	Very few videos	✓	Usually one	Creator pays
Photo describers	One photo at a time	✗	Many	Free
AI video summaries	Any video, after the fact	✗	Many	Varies
Enjoyable Pro	**Whole films, preprocessed**	✓	**130+**	**≈ US$0.08**
Enjoyable Lite	**Any YouTube video, in real time**	✓	**33**	**≈ US$0.016**

In short: **900× cheaper** · **33 languages** · **dubs foreign dialogue**

List of features offered by the solution



Any YouTube link

Paste a link or pick a ready video. **No account, no install.**

Fits the pauses

Each line is timed and paced to **fit the gap** in the dialogue.

Voice-over (dubbing)

Speech in another language is **spoken again in yours**, in a second voice.

“What's on screen?”

Ctrl+Shift+S describes the current moment **on demand**.

33 languages

Japanese, Korean, Chinese, Hindi, Indonesian, Malay, Vietnamese and more.

Keyboard first

Every action has a shortcut; spoken confirmations; WCAG 2.2 targets.

Low-vision mode

Large, high-contrast captions at the top of the video.

Voice on this device

Optional: the voice **runs in the browser** with WebGPU, after one download.

Full edition: the desktop app describes videos on any website and can run fully offline with Gemma.

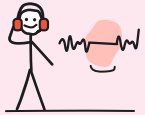
Process flow

What happens after a viewer pastes a link



01 Paste a link

The page wakes the **Cloud Run** service and opens the YouTube player.



02 Find pauses

Gemini listens to the next 15–60 s: **when people speak**, and what they say.



03 Write lines

Gemini watches the same stretch and writes **one short line per pause**.



04 Voice them

Local voices speak each line, **fitted to its pause**.



05 Deliver live

Firestore and **Cloud Storage** send each line to the page **at once**.

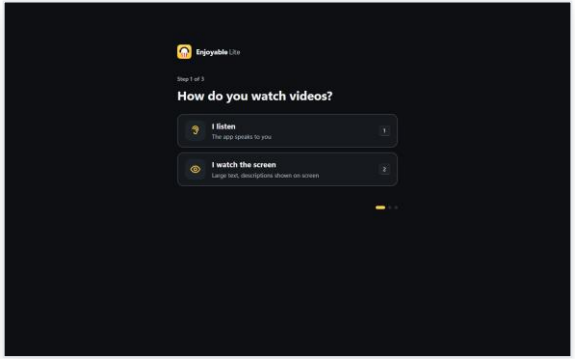


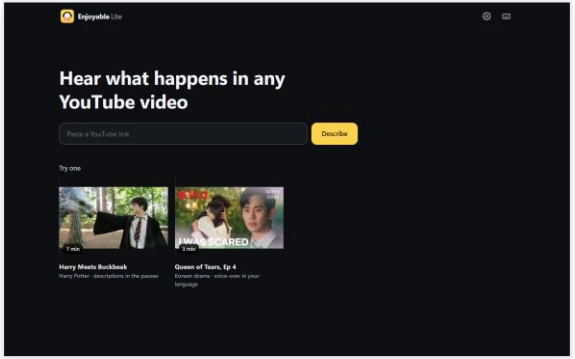
06 Hear it

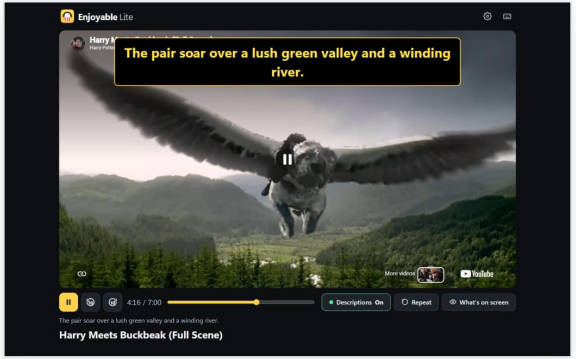
Each line plays **in its pause**; the video sound dips underneath.

Wireframes / screens

Three screens: a short welcome, a home page with one field, and the player







1 Welcome

One question per screen, answered with a number key

2 Home

Paste a link, or pick a ready video

3 Watch

Big controls, the description on screen, every action a shortcut

Made for blind and low-vision viewers: dark and high contrast (text ≥ 7:1).

Architecture diagram of the proposed solution



Viewer's browser

YouTube player · keyboard and screen reader

Gemini

3.8 Flash listens, writes and translates

watches

YouTube

The video itself

Firebase Hosting

Cloud Run

Prepares each video ahead of the viewer; speaks with local voices

live

Firestore

Lines, live to the page

Cloud Storage

Voice clips

voice clips

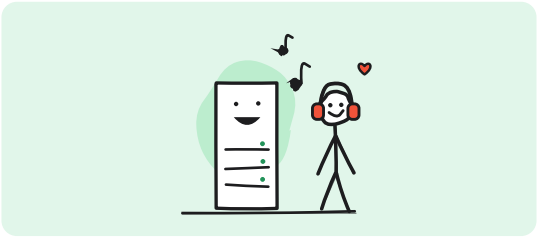
Technologies used in the solution



Google Cloud

Hosting, compute, data and AI

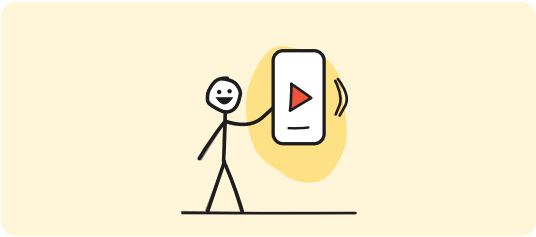
- Gemini 3.8 Flash
- Cloud Run
- Cloud Firestore
- Cloud Storage
- Firebase Hosting
- Secret Manager
- Cloud Build



Open-source voices

Run inside Cloud Run, no per-word cost

- Supertonic 3 · 31 languages
- Kokoro · Chinese
- ONNX Runtime
- ffmpeg · audio



Web app

Any browser, no install, no login

- HTML + JavaScript
- YouTube IFrame Player
- Firebase JS SDK
- Kokoro in the browser · WebGPU
- WCAG 2.2

Estimated implementation cost

Measured on the prototype

One minute of a new video

Gemini
≈ US$0.016 a minute, dubbing included

Cloud Run
Free tier, then ≈ US$0.16 an hour

Firestore, Storage
Free tier at this scale

Already described
US$0: it is read, not made again

Total ≈ US$0.016



US$15–75

Professional description



≈ US$0.016

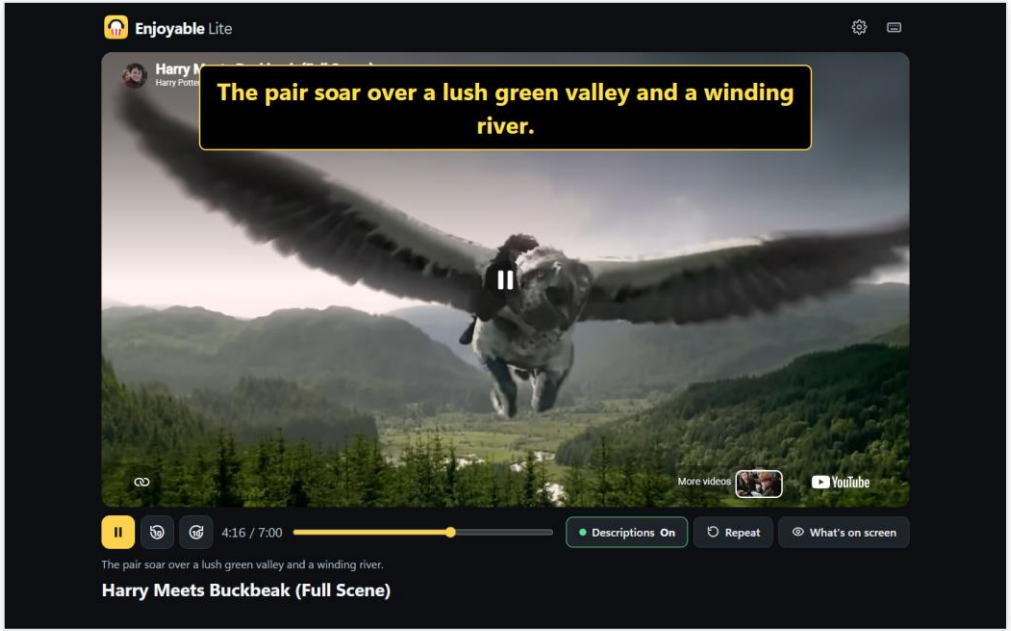
Enjoyable Lite

900× cheaper

Snapshots of the prototype

https://enjoyable-lite.ables.com.my/



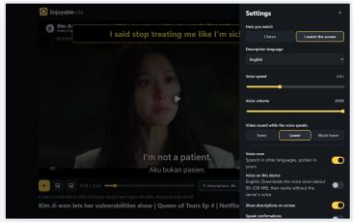


Harry Potter: **descriptions played in the gaps** between the dialogue



Korean drama

Dialogue **dubbed into English**, in a second voice



Settings

33 languages, voice speed, mixing, captions



Shortcuts

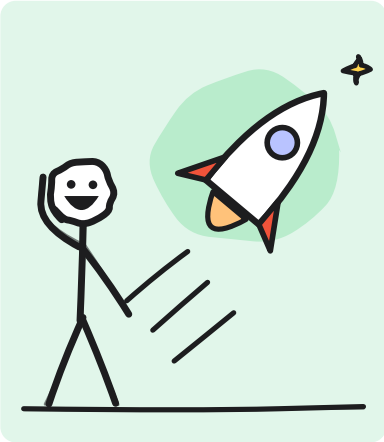
Every action from the keyboard

Prototype performance / benchmarking



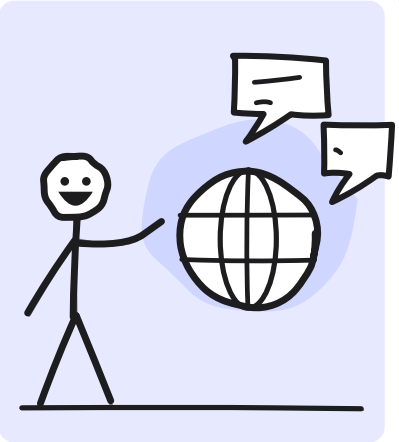
< 1 min

until a **new video** plays straight through, live on Cloud Run



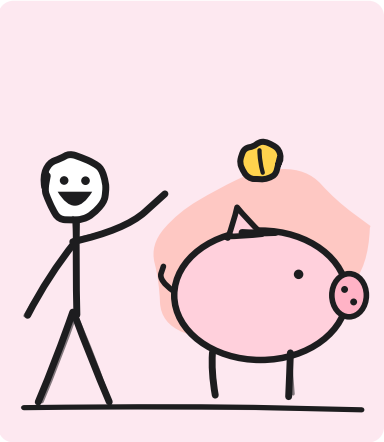
1.5×

faster than real time: 60 s of video ready in about 40 s



33

languages with a natural voice, **plus dubbing**



≈ US$0.016

per minute of new video, **free after that**

Additional details / future development



PRIVACY

Fully on-device

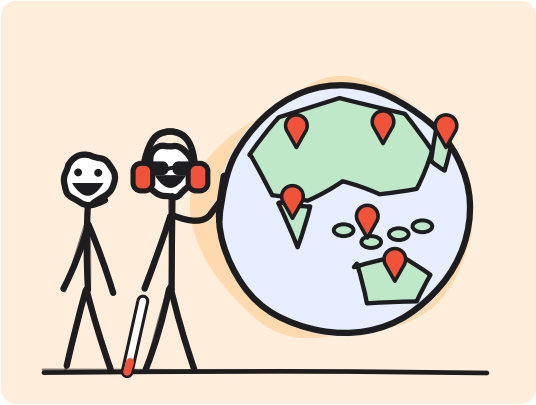
The desktop app runs **every model on the laptop**, the vision model too: private, and offline.



REACH

Any streaming site

A browser extension for **Tonton, iQIYI and more**.



COMMUNITY

Pilots across JAPAC

With blind associations **across Asia-Pacific**, in their own languages.

Links

Try the prototype

enjoyable-lite.ables.com.my

GitHub public repository

github.com/yayadontknow/enjoyable-lite

Demo video

Watch on Google Drive

ABLES

Here, we **make dreams**, we **sell dreams**.

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Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

